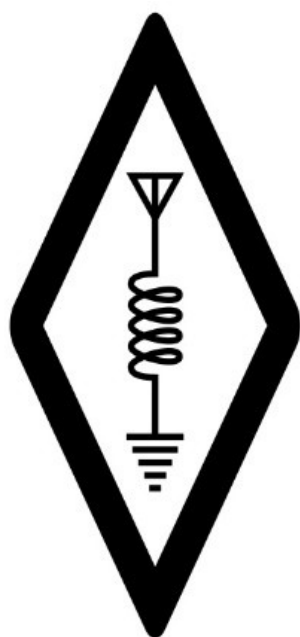




2E0UMR

Communication Reference Book

Version 2 Revision 1

GB

(September 2026)

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Ham Operator Information

Callsign:

QTH:

QTH Locator:

DMR ID:

NXDN ID:

APRS-IS Passcode:

Winlink Email:

Phonetic Alphabets

Alpha **B**ravo **C**harlie **D**elta **E**cho **F**oxtrot
Golf **H**otel **I**ndia **J**uliet **K**ilo **L**ima **M**ike
November **O**scar **P**apa **Q**uebec **R**omeo
Sierra **T**ango **U**niform **V**ictor **W**hiskey **X**-ray
Yankee **Z**ulu

Morse Code

A	B	C	D	E
. -	- . . .	- . - .	- . .	.
F	G	H	I	J
. . - .	- - .		. .	. - - -
L	M	N	O	P
. - . .	- -	- .	- - -	. - - .
Q	R	S	T	U
- - . -	. - .		-	. . -
V	W	X	Y	Z
. . . -	. - -	- . . -	- . - -	- - . .
1	2	3	4	5
. - - - -	. . - - -	. . . - -	 -	
6	7	8	9	0
-	- - . . .	- - - . .	- - - - .	- - - - -

Mobile Radio Operation Checklist

- HF/ VU Transceiver []
- Power source and power cables []
- Appropriate antenna and feed line []
- Logbook []
- Tools []
- Safety signs []
- First aid kit []
- Additional radios (if any) []
- PC / Laptop / Tablet and Interface (if using data mode) []

Notes

Notes

Emergency Message Category

Routine- Normal message

Priority- Urgent message

Immediate- Extremely urgent message

Emergency- Life at risk message

Emergency & non-emergency Phone numbers (UK)

- **999 or 112**

(Ambulance/Fire/Police/Coast guard/SAR)

- **101** (Non-Emergency Police)
- **111** (Non-Emergency NHS)
- **0800 111 999**

(National Gas Emergency)

- **105 or 0800 6783 105**

(National Power Grid)

RST Report reference

Strength

1. Faint—signals barely perceptible
2. Very weak signals
3. Weak signals
4. Fair signals
5. Fairly good signals
6. Good signals
7. Moderately strong signals
8. Strong signals
9. Extremely strong signals

Readability

1. Unreadable
2. Barely readable, occasional words distinguishable
3. Readable with considerable effort
4. Readable with practically no effort
5. Perfectly readable

Q Codes

QRG Your exact frequency (or that of _____) is _____ kHz.

Will you tell me my exact frequency (or that of _____)?

QRL I am busy (or I am busy with _____).
Are you busy?

Usually used to see if a frequency is busy.

QRM Your transmission is being interfered with _____

(1. Nil; 2. Slightly; 3. Moderately; 4. Severely; 5. Extremely.)

Is my transmission being interfered with?

QRN I am troubled by static _____. (1 to 5 as under QRM.)

Are you troubled by static?

QRO Increase power. Shall I increase power?

QRP Decrease power. Shall I decrease power?

Q Codes

QRQ Send faster (_____wpm). Shall I send faster?

QRS Send more slowly (_____wpm). Shall I send more slowly?

QRT Stop sending. Shall I stop sending?

QRU I have nothing for you. Have you anything for me?

QRV I am ready. Are you ready?

QRX I will call you again at _____hours (on _____kHz).

When will you call me again? Minutes are usually implied rather than hours.

QRZ You are being called by _____ (on _____kHz).

Who is calling me?

QSB Your signals are fading. Are my signals fading?

Q Codes

QSK I can hear you between signals; break in on my transmission.

Can you hear me between your signals and if so can I break in on your transmission?

QSL I am acknowledging receipt.

Can you acknowledge receipt (of a message or transmission)?

QSO I can communicate with _____ direct (or relay through _____).

Can you communicate with _____ direct or by relay?

QSP I will relay to _____. Will you relay to _____?

QST General call preceding a message addressed to all amateurs.

QSX I am listening to _____ on _____ kHz. Will you listen to _____ on _____ kHz?

QSY Change to transmission on another frequency (or on _____ kHz).

Q Codes

Shall I change to transmission on another frequency (or on _____kHz)?

QTC I have _____messages for you (or for _____).

How many messages have you to send?

QTH My location is _____. What is your location?

QTR The time is _____. What is the correct time?

Emergency Communication Abbreviations/Prowords

ACKNOWLEDGE Let me know that you have received and understood this message.

AFFIRM Yes.

ALL AFTER/BEFORE Indicates part of a message

APPROVED Permission is granted for the proposed action.

BREAK Indicates the separation between messages.

CANCEL Annul the previously transmitted instruction.

CLOSE DOWN Cease operation of the radio station.

CONFIRM Did you correctly receive this message?

CORRECT That is correct

Emergency Communication Abbreviations/Prowords

CORRECTION An error has been made in MY transmission. The correct version is...

DECIMAL Decimal Point.

DIRECT CONTACT Contact without repeater.

DISREGARD Consider that transmission as not sent.

EMERGENCY I have a message of life and death urgency.

I SAY AGAIN I Repeat my message again.

IMMEDIATE I have a message of life and death urgency.

MESSAGE I have a message for you.

Emergency Communication Abbreviations/Prowords

MESSAGE ENDS The entire text of the message has been sent.

MISTAKE Error in the transmission.

NEGATIVE No.

OVER My transmission is ended. Expecting response.

OUT My Transmission ended, not expecting response.

PASS YOUR MESSAGE Proceed with your message.

PRIORITY I need to interrupt with a PRIORITY message.

RADIO CHECK Report the readability of my transmission.

READ BACK Repeat your message.

RELEVANT I have a message relevant to the current exchange of transmissions.

Emergency Communication Abbreviations/Prowords

REPORT Pass requested information.

REQUEST I wish to take certain action.

ROGER I have received and understood your message.

ROGER SO FAR I have received the message so far.

SAY AGAIN Repeat all.

SEND Proceed with your message.

SITREP Situation Report.

STAND BY Wait I will call you.

TALK-THROUGH Contact via repeater.

WAIT Do not transmit until I call you.

WAIT ONE Do not transmit until I call you in one minute.

WILCO Will comply with the instructions.

YES-YES Yes.

Emergency Comms Frequencies

HF

- 7.110 MHz LSB Global
- 14.300 MHz USB Global
- 18.160 MHz USB Global
- 21.360 MHz USB Global
- 50.210MHz NFM UK

Aviation (RX Only)

- 121.500 MHz AM

VHF

- 144.260 MHz USB (UK)
- 144.625-144.775 MHz NFM (UK)

Marine (Restrictions apply)

- 156.800 MHz FM “CH 16”

UHF

- 438.400 MHz (-7.6MHz) input FM (RAYNET)

UK 70 cm Band

- 430.0125 to 430.0750 Internet Voice gateway (12.5kHz Steps)
- 430.800 EMG 7.6MHz Talk through
- 431.075 to 431.1750 Internet Voice gateway (25kHz steps)
- 432.00 to 432.0250 Moon bounce
- 432.200 SSB Centre of activity
- 432.400 to 432.500 Beacons Exclusive
- 432.500 NB SSTV
- 432.6250 to 432.6750 Digital Communication (25kHz steps)
- 432.700 FAX Centre of activity
- 432.7750 EMG 1.6MHz Talk through
- 432.800 to 432.990 Beacons
- 432.400 Region1 SSTV
- 433.400-433.5750 FM 25K spacing 433.500 calling
- 433.800 to 434.5940 Digital Communication (12.5kHz steps)
- 434.3750 EMG 1.6MHz Talk through
- 434.4750 to 434.5250 Internet Voice gateway (25kHz steps)
- 435.000 to 438.00 Satellites and Fast Scan TV
- 438.000 to 440.00 IARU Region 1 Digital
- 438.400 MHz -7.6MHz input FM (Raynet)
- 439.9875 POCSAG Pager
- 439.9125 LoRa APRS (BW: 125kHz, SF: 12, CR: 4/5 (enter as 5))

UK 70 cm Simplex (FM) frequency

- 433.400 MHz U272
- 433.4250 MHz U274
- 433.4500 MHz U276
- 433.4750 MHz U278
- 433.500 MHz U280 **Calling**
- 433.5250 MHz U282
- 433.550 MHz U284
- 433.575 MHz U286

UK 70cm DMR Simplex frequency (TS: 1, TG: 9, CC: 1)

- DH1: 438.5875 MHz
- DH2: 438.6000 MHz
- DH3: 438.6125 MHz **DMR Calling**
- DH4: 438.6250 MHz
- DH5: 438.6375 MHz
- DH6: 438.6500 MHz
- DH7: 438.6625 MHz
- DH8: 438.6750 MHz

UK TETRA EXPERIMENTAL DMO SIMPLEX

Encryption: Clear, **ISSI:** Your DMR ID, **GSSI/TG:** 10, **MCC:** 753, **MNC:** 2361, **Frequency:** 430.3125 MHz

UK 2M Band

- 144.00 to 144.110 EME exclusive
- 144.050 CW calling
- 144.110 to 144.150 PSK31 centre of activity
- 144.150-144.400 SSB 144.300 SSB calling
- 144.175 Microwave Talk back
- 144.250 GB2RS news and Slow Morse
- 144.260 USB Emergency
- 144.300 SSB Calling
- 144.360 to 144.399 SSB, CW
- 144.400 to 144.490 Beacons only
- 144.500 to 144.794 All modes
- 144.500 SSTV/FAX
- 144.525 ATV SSB talk back
- 144.600 RTTY (FSK)
- 144.750 ATV Talk back
- 144.775 to 144.794 EMG
- 144.800 APRS
- 144.8250 High Speed Packet Users
- 144.8625 BBS and nodes
- 145.200 Space Comms Earth to Space
- 145.250 Slow Morse
- 145.300 Local RTTY
- 145.200 to 145.5875 FM 12.5Khz spacing 145.500 calling
- 145.5250 GB2RS News
- 145.800 ISS SSTV Downlink Space to Earth

UK 2M Simplex (12.5Khz) Frequencies

- 145.2000 MHz V16
- 145.2125 MHz V17
- 145.2250 MHz V18
- 145.2375 MHz V19
- 145.2500 MHz V20
- 145.2625 MHz V21
- 145.2750 MHz V22
- 145.2875 MHz V23
- 145.3000 MHz V24
- 145.3125 MHz V25
- 145.3250 MHz V26
- 145.3375 MHz V27
- 145.3500 MHz V28
- 145.3625 MHz V29
- 145.3750 MHz V30
- 145.3875 MHz V31
- 145.4000 MHz V32
- 145.4125 MHz V33
- 145.4250 MHz V34
- 145.4375 MHz V35
- 145.4500 MHz V36
- 145.4625 MHz V37
- 145.4750 MHz V38
- 145.4875 MHz V39
- 145.5000 MHz V40 **FM Calling**
- 145.5125 MHz V41
- 145.5250 MHz V42 **GB2RS News**

UK 2M Simplex (12.5Khz) Frequencies

- 145.5375 MHz V43
- 145.5500 MHz V44 **Rally/Exhibition Talk-in**
- 145.5625 MHz V45
- 145.5750 MHz V46
- 145.5875 MHz V47

UK 6M Band

- 50.000 To 50.100 CW
- 50.000 to 50.080 Beacons
- 50.090 CW Centre of activity
- 50.100 to 50.500 NB Mode
- 50.110 DX Call
- 50.150 SSB centre of activity
- 50.210MHz NFM Emergency
- 50.255 JT44
- 50.250 PSK31 Centre of activity
- 50.260 to 50.280 PSK 441
- 50.270 FSK 441
- 50.500-52.000 All Mode
- 50.510 SSTV (AFSK)
- 50.550 FAX
- 51.430 to 51.590 FM
- 51.530 GB2RS News
- 51.950 to 51.990 may be used for emergency

UK 10M Band

- 28.00 to 28.070 CW
- 28.055 QRS (Slow CW) Centre of activity
- 28.060 QRP Centre of activity
- 28.680 SSTV
- 28.070 to 28.190 Data mode
- 28.190 to 28.199 IBP- Regional Shared Beacons
- 28.199 to 28.200 IBP- WW Time Shared Beacons
- 28.201 to 28.225 IBP-Continuous Duty Beacons
- 28.225 to 28.300 All mode Beacons
- 28.300 to 28.320 Data mode
- 28.300 to 29.000 QRP SSB 28.360 Centre of activity
- 29.520 to 29.550 FM Simplex 10kHz Ch
- 29.600 FM calling
- 29.610 to 29.6500 FM Simplex 10kHz Ch

UK 12M Band

- 24.890 to 24.915 CW
- 24.906 QRP Centre of activity
- 24.915 to 24.925 NB mode
- 24.929 to 24.931 IBP Beacons
- 24.940 to 24.990 All mode

UK 15M Band

- 21.000-21.070 CW
- 21.055 QRS (Slow Morse)
- 21.060 QRP
- 21.070 to 21.090 NB Mode
- 21.090 to 21.110 NB Data mode
- 21.110 to 21.120 Data mode
- 21.120 to 21.149 NB Mode
- 21.149 to 21.151 IBP Beacons
- 21.151 to 21.450 All Mode
- 21.340 SSTV
- 21.360 Global EMG

UK 17M Band

- 18.068 to 18.095 CW
- 18.096 QRP CW
- 18.095 to 18.105 NB Mode
- 18.105 to 18.109 NB Data mode
- 18.109 to 18.111 Beacons
- 18.111 to 18.120 Data mode
- 18.120 to 18.168 All Mode
- 18.160 Global EMG

UK 20M Band

- 14.000 to 14.060 CW Contest
- 14.060 to 14.070 CW
- 14.060 QRP CW
- 14.099 to 14.101 Beacons
- 14.101 to 14.112 Data mode
- 14.112 to 14.125 All mode
- 14.125 to 14.300 SSB contest
- 14.195 Priority for DX
- 14.230 SSTV
- 14.285 QRP SSB
- 14.300 to 14.350 All mode
- 14.300 Global EMG

UK 30M Band QRP

- 10.100 to 10.140 CW
- 10.116 QRP
- 10.140 to 10.150 NB

UK 40M Band

- 7.000 to 7.035 CW
- 7.030 CW QRP
- 7.035 to 7.038 NB mode
- 7.038 to 7.040 NB Data mode
- 7.040-7.043 Data mode
- 7.043 SSTV
- 7.043-7.200 All mode
- 7.060 Region 1 Centre of activity
- 7.110 Global EMG

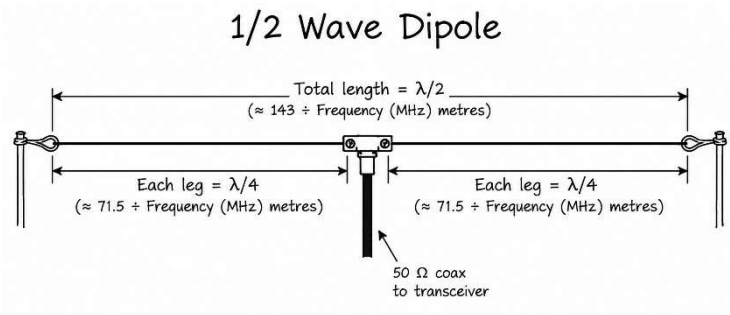
HF FT-8 Frequencies

- 40m: 7.074 MHz
- 30m: 10.136 MHz
- 20m: 14.074 MHz
- 17m: 18.100 MHz
- 15m: 21.074 MHz
- 12m: 24.915 MHz
- 10m: 28.074 MHz

VHF/ UHF FT-8 Frequencies

- 6m: 50.313 MHz or 50.323 MHz
- 2m: 144.174 MHz

1/2 Wave Dipole Dimensions (Total length)



- | | |
|---------|----------|
| • 70 cm | ~34.3 cm |
| • 2 m | ~98 cm |
| • 6 m | ~2.86 m |
| • 10 m | ~5 m |
| • 20 m | ~10.1 m |
| • 40 m | ~20.4 m |

Appendix

SSB Single Side Band.

LSB Lower Side Band (used below 10MHz).

USB Upper Side Band (used above 10MHz).

EMG Emergency.

NB Narrow Band.

CW Continuous Wave/Morse Code.

TS Time Slot

CC Colour Code

TG Talk Group

BW Bandwidth

SF Spreading Factor

CR Coding Rate



Sources

<https://www.raynet-uk.net/manual.asp>

<https://rsgb.org/main/operating/band-plans/>

<https://www.arrl.org/files/file/Get%20on%20the%20Air/Comm%20w%20Other%20Hams-Q%20Signals.pdf>